

ORIGINAL RESEARCH

Genetically Mediated Differences in LDL Cholesterol and Risk of Venous Thromboembolism

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ABSTRACT

BACKGROUND Venous thromboembolism (VTE) is a common and clinically significant cardiovascular condition. Although low-density lipoprotein (LDL) cholesterol is causally linked to atherosclerotic disease, its role in VTE remains unclear. Meta-analyses of randomized trials have suggested modest reductions in VTE with lipid-lowering therapy, but whether these effects are LDL mediated is uncertain.

OBJECTIVES This study sought to evaluate whether lifelong genetically mediated differences in low-density lipoprotein cholesterol (LDL-C) are associated with VTE risk.

METHODS Whole-genome sequencing data from 430,049 UK Biobank and 283,609 All of Us participants were analyzed. Rare protein-truncating and AlphaMissense-predicted damaging missense variants (minor allele frequency <0.1%) in *LDLR*, *APOB*, and *PCSK9*, as well as the common *PCSK9* R46L variant, were evaluated. Associations with LDL-C were assessed using linear regression, and VTE associations using Firth logistic regression with fixed-effect meta-analysis. Incident VTE was evaluated using Cox models.

RESULTS Rare functional variants in *LDLR*, *APOB*, and *PCSK9* produced substantial gene-specific differences in LDL-C (up to ~50% differences). Despite these marked lifelong differences, rare variant burdens were not consistently associated with VTE across cohorts, although smaller effect sizes could not be excluded. Meta-analyzed ORs ranged from 0.60 to 1.07. Sensitivity and prospective analyses yielded similar findings. Complementary common variant and multivariable Mendelian randomization analyses suggested that previously reported LDL-C associations with VTE may partly reflect broader metabolic and obesity-related pathways. Positive control analyses of established thrombophilia variants demonstrated expected associations.

CONCLUSIONS Across 2 large biobanks, genetically mediated differences in LDL-C were not consistently associated with VTE risk, suggesting that genetically mediated differences in LDL-C are unlikely to exert a large effect on venous thrombosis risk, although small effects cannot be excluded. Complementary common variant analyses further suggest that previously reported LDL-C associations with VTE may partly reflect broader metabolic and obesity-related pathways rather than LDL-C itself. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS
AND ACRONYMS

AS = aortic stenosis

ASCVD = atherosclerotic
cardiovascular disease

CAD = coronary artery disease

LDL = low-density lipoprotein

LDL-C = low-density
lipoprotein cholesterolVTE = venous
thromboembolism

Venous thromboembolism (VTE), comprising deep vein thrombosis and pulmonary embolism, is a major cause of cardiovascular morbidity and mortality worldwide. Its lifetime risk approaches 8%,¹ and recurrence remains substantial despite guideline-directed anticoagulation.² Clinical risk assessment is dominated by acquired factors such as surgery, immobilization, trauma, cancer, and hormonal therapy¹; yet, the biological determinants of VTE are considerably less well-defined compared with atherosclerotic cardiovascular disease (ASCVD).² In ASCVD, low-density lipoprotein cholesterol (LDL-C) is a well-established causal factor and therapeutic target, whereas the relevance of low-density lipoprotein (LDL)-related pathways to VTE remains uncertain.³

Recent clinical evidence has renewed interest in this question. A meta-analysis reported that lipid-lowering therapies, particularly high-intensity statins and combination LDL-lowering regimens, were associated with a modest reduction in incident VTE.⁴ These findings, observed across randomized trials and observational cohorts, raised the possibility that LDL lowering, or therapy-specific pleiotropy, may exert protective effects on venous thrombosis.^{4,5} Primary prevention of VTE remains limited beyond anticoagulation in high-risk settings.⁶ If LDL-C plays a causal role in venous thrombosis, this would identify a broadly scalable and well-characterized therapeutic pathway with extensive safety data. Clarifying whether LDL lowering per se modifies VTE risk therefore has important clinical implications.

Human genetics provides a unique opportunity to disentangle LDL-mediated effects from potential additional drug-specific mechanisms. Rare protein-truncating or function-disrupting variants in *LDLR*, *APOB*, and *PCSK9* produce lifelong, large perturbations in LDL-C and have been critical in establishing the causal role of LDL in coronary artery disease (CAD).⁷ Likewise, the common *PCSK9* missense variant R46L produces moderate LDL reductions and confers strong protection against ASCVD.⁸ In addition to CAD, rare LDL-altering variants in *LDLR*, *APOB*, and *PCSK9* have been shown to strongly influence aortic stenosis (AS) risk, reflecting the cumulative effects of lifelong LDL exposure and providing a genetically validated comparator phenotype.⁹ If lifelong genetically mediated LDL lowering reduces VTE risk, as suggested by pharmacologic studies,⁴ carriers of these variants would be expected to exhibit lower rates of VTE.

To address this question, we leveraged large-scale whole-genome sequencing (WGS) data from the UK Biobank (UKB) and the All of Us (AoU) Research Program to systematically evaluate whether rare LDL-modifying variants and the common *PCSK9* R46L variant are associated with VTE risk, with CAD and AS analyzed as comparator cardiovascular outcomes. By using genetics as a natural experiment, we aimed to clarify whether LDL-C plays a causal role in VTE and to inform the clinical relevance of lipid pathways in venous thrombotic disease.

METHODS

UK BIOBANK. UKB is a large, population-based prospective cohort that enrolled >500,000 individuals

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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aged 40 to 69 years across the United Kingdom between 2006 and 2010.¹⁰ At baseline, participants provided written informed consent, completed detailed questionnaires, and underwent extensive phenotyping, including anthropometric measurements. Additional data sources included medical interviews, electronic health records (EHRs), and national death registries.^{10,11}

ALL OF US. The AoU Research Program,¹² launched in 2018¹³ by the U.S. National Institutes of Health, is a diverse longitudinal cohort with the goal of enrolling 1 million participants aged ≥ 18 years. The program prioritizes inclusion of individuals historically under-represented in biomedical research. To date, >800,000 participants have consented, with >560,000 completing initial enrollment, including health questionnaires and biospecimen collection. Phenotypic data are collected from multiple sources, including participant-reported information and EHR linkage,¹⁴ which includes International Classification of Diseases-Ninth/Tenth Revision, SNOMED, and CPT codes. In the February 2025 version 8 data release, WGS was completed for approximately 414,000 participants using Illumina NovaSeq 6000 machines. Library preparation was standardized using PCR-free Kapa HyperPrep, and variant calling was performed using DRAGEN version 3.7.8 across all Genome Centers. Centralized quality control (QC) was conducted as described in the AoU Genomic Quality Report.¹⁵

QC FOR WGS IN UKB AND AoU. In UKB, WGS data for >490,000 participants was released to the DNAnexus platform. We used pVCF files released in 2024, processed through the DRAGEN pipeline. We processed the >150,000 pVCF files through various steps, which included genotype-, variant-, and sample-level QC procedures. Filtering details can be found in the [Supplemental Methods](#). Analyses were conducted under approved application number 17488.

In AoU, we obtained genotypes in PLINK¹⁶ binary format from the jointly called exome call set (version 8) provided by AoU. This data set includes only high-quality genotypes, with multiallelic variants already split into biallelic variants and variants with allele frequency = 0 removed, as described in the AoU Genomic Quality Report.¹⁵ We further filtered variants and samples as can be found in the [Supplemental Methods](#). Use of data from the AoU Research Program was approved under a data use agreement between Massachusetts General Hospital and the AoU Research Program.

VARIANT ANNOTATION. In both data sets, genetic variants were annotated using the Variant Effect

Predictor (VEP version 105)¹⁷ with the LOFTEE¹⁸ plugin, alongside annotations from the dbNSFP¹⁹ database version 4.2. Predicted truncating variants (tv) in *LDLR*, *APOB*, and *PCSK9* were identified using LOFTEE,¹⁸ excluding those variants that were annotated by LOFTEE as low confidence or flagged based on characteristics associated with increased false-positive rates. We also identified likely damaging missense variants in the same genes by using predictions from AlphaMissense.²⁰ We downloaded the AlphaMissense predictions,²⁰ using the AlphaMissense_hg38.tsv.gz file (August 2023 release). Because transcript IDs used by the developers did not always match ENSEMBL canonical transcripts, we mapped AlphaMissense²⁰ predictions to canonical transcripts using ENSEMBL gene IDs based on GENCODE²¹ version 45. Variants labeled as “predicted_pathogenic” classification by AlphaMissense were classified as predicted damaging missense (AM) variants. To decrease the number of false positives and to identify true high-impact variants, both variant categories were limited to rare variants only (defined as a maximum minor allele frequency <0.1%). Detailed variant-level annotations for rare variants identified in the AoU cohort and included in *LDLR*, *APOB*, and *PCSK9* burden analyses are provided in [Supplemental Tables 1 to 3](#). These annotations include predicted functional consequence, ClinVar annotation, and inferred direction of effect on LDL-C.

OUTCOME DEFINITION. VTE was defined using harmonized diagnostic and procedural criteria across UKB and AoU, integrating self-reported data, inpatient records, International Classification of Diseases-9th/10th Revision codes, SNOMED-CT concepts, and VTE-related procedures.^{22,23} Codes related to superficial phlebitis, portal vein thrombosis, and coagulation disorders were excluded. Individuals with ≥ 1 qualifying VTE code were considered cases. CAD and AS were analyzed as comparator cardiovascular outcomes, given their well-established genetic associations with lifelong LDL-C exposure. Full phenotype definitions for VTE, CAD, and AS in UKB and AoU are provided in [Supplemental Tables 4 and 5](#). To evaluate the robustness of VTE ascertainment, we additionally performed sensitivity analyses using alternative VTE phenotype definitions derived from previously published and validated algorithms.^{24,25} Effect estimates were compared across definitions to assess the potential impact of phenotype misclassification on the observed associations.

STATISTICAL POWER ANALYSIS. Statistical power to detect associations among *APOB*, *PCSK9* carrier status, and VTE was estimated using a Wald test under

a logistic regression framework, assuming asymptotic normality. Power was calculated at a 2-sided $\alpha = 0.05$ across a range of assumed ORs (0.1-1.0), using observed case proportion and carrier frequencies for each variant class. The OR required to achieve 80% power (OR₈₀) was determined numerically.

POSITIVE CONTROL ANALYSIS. As a positive control for the rare variant analytic framework, we replicated the classification of classic thrombophilia variants described by Manderstedt et al.²⁶ High-risk variants were defined as nonsynonymous variants annotated as disease-causing in the Human Gene Mutation Database (HGMD) within the anticoagulant genes *SERPINC1*, *PROC*, and *PROS1*. Low-risk variants were defined as non-HGMD nonsynonymous variants in the same genes. In addition, 2 established common thrombophilia variants, factor V Leiden (rs6025) and prothrombin G20210A (rs1799963), were analyzed. The variants list used in UKB and AoU can be found in [Supplemental Tables 6 and 7](#). Genotypes were coded as noncarrier, heterozygote, or homozygote and modeled using logistic regression with noncarriers as the reference group. Replication of previously reported effect sizes served as validation of variant annotation and analytical procedures.

MULTIVARIABLE MENDELIAN RANDOMIZATION ANALYSES. To further integrate rare variant findings with common variant lipid genetic evidence, we performed complementary multivariable Mendelian randomization (MR) and polygenic risk score (PRS) analyses from the AoU cohort.

Two multivariable MR models were implemented using inverse variance-weighted estimation in European ancestry participants. The first model replicated the framework previously described by Klarin et al.²⁴ using the 222-variant lipid genetic instrument, jointly modeling LDL-C, high-density lipoprotein cholesterol (HDL-C), and triglycerides as exposures. Harmonized SNP-level association estimates used in these analyses are provided in [Supplemental Table 8](#). Variant VTE associations in AoU European ancestry participants were estimated using logistic regression adjusted for age, age², sex, and the first 10 principal components by PLINK.¹⁶

In the second model, LDL-C and body mass index (BMI) were jointly modeled to evaluate whether obesity-related pathways might contribute to previously reported common variant LDL-C associations with VTE. LDL-C instrument summary statistics were obtained from the Global Lipids Genetics Consortium exome array study,²⁷ and BMI instrument summary statistics were obtained from the GIANT Consortium European ancestry meta-analysis.²⁸ Harmonized

SNP-level association estimates used in these analyses are provided in [Supplemental Table 9](#).

Additionally, we constructed an LDL-C PRS using the LDpred2 method, based on LDL-C GWAS summary statistics from the Global Lipids Genetics Consortium²⁹ excluding UKB participants. Individuals were stratified into quintiles of LDL-C PRS, and baseline clinical and metabolic characteristics were compared between the top quintile and rare coding variant carriers using standardized mean differences, 2-sided Student's *t*-tests for continuous variables, and chi-square tests for categorical variables.

STATISTICAL ANALYSIS. Associations between variant carrier status and LDL-C were assessed using linear regression adjusted for age, age², sex, genotyping array (UKB only), and the first 10 principal components. LDL-C values for participants on lipid-lowering therapy were corrected by dividing by 0.7 as previously described.^{29,30}

Disease associations were evaluated using Firth logistic regression, with VTE, CAD, or AS as outcomes and the same covariates. Participants with either prevalent or incident VTE (or correspondingly CAD or AS) were included as cases. SEs and corresponding Wald 95% CIs were calculated based on the *P* values and effect estimates from Firth logistic regression. Results from UKB and AoU were combined via inverse-variance-weighted fixed-effect meta-analysis. Heterogeneity was assessed using Cochran's Q and I² statistics.

To account for potential diagnostic ascertainment in the context of CAD, we performed additional stringent sensitivity analyses in UKB and AoU by evaluating the associations of each variant category with VTE after excluding all participants with CAD.

Finally, we conducted analyses of the incidence of VTE among UKB participants who were free of VTE at study enrollment, using Cox proportional hazards regression with sex, age, age², the first 10 principal components, and genotyping array as covariates. In the AoU cohort, incident VTE analyses were not performed caused by the relatively short follow-up duration since cohort initiation and potential variability in event ascertainment across sites.

All statistical tests were two-sided, and all analyses were conducted in R (The R Foundation for Statistical Computing, Vienna, Austria) unless otherwise specified.

RESULTS

CHARACTERISTICS OF STUDY PARTICIPANTS. After sample QC, 430,049 unrelated participants from UKB (46.1% men; 21,048 with VTE) ([Table 1](#)) and 283,609

TABLE 1 Characteristics of the UK Biobank Study Participants by Rare Variant Carrier Status at Study Enrollment

	Noncarriers (n = 412,470)	LDLR (n = 735)	APOB (n = 1,614)	PCSK9 Rare (n = 1,151)
Age at recruitment, y	57.03 ± 8.07	56.68 ± 8.34	56.50 ± 8.14	56.83 ± 8.30
Female	222,158 (53.86)	416 (56.60)	903 (55.95)	655 (56.91)
BMI, kg/m ²	27.41 ± 4.78	27.09 ± 4.74	27.72 ± 4.88	27.28 ± 4.68
Lipid measurements				
Uncorrected LDL-C, mg/dL	138.05 ± 33.41	168.05 ± 53.04	106.88 ± 39.60	111.05 ± 29.93
Corrected LDL-C, mg/dL	147.98 ± 35.79	206.51 ± 61.83	112.84 ± 44.10	114.02 ± 30.00
HDL-C, mg/dL	56.04 ± 14.80	55.66 ± 13.63	59.01 ± 16.99	57.35 ± 15.70
Triglycerides, mg/dL	154.91 ± 90.53	139.70 ± 87.45	117.00 ± 79.03	145.12 ± 83.95
Total cholesterol, mg/dL	220.76 ± 44.04	254.47 ± 67.40	184.03 ± 50.56	189.70 ± 40.43
Disease				
Venous thromboembolism	20,123 (4.88)	37 (5.03)	76 (4.71)	52 (4.52)
Coronary artery disease	25,088 (6.08)	104 (14.15)	67 (4.15)	34 (2.95)
Aortic stenosis	5,498 (1.33)	36 (4.90)	16 (0.99)	8 (0.70)
Hypertension	168,463 (40.84)	307 (41.77)	707 (43.80)	452 (39.27)
Type 1 diabetes	4,289 (1.04)	6 (0.82)	23 (1.43)	14 (1.22)
Type 2 diabetes	37,236 (9.03)	74 (10.07)	192 (11.90)	119 (10.34)
Self-reported ancestry				
Asian or Asian British	8,663 (2.11)	21 (2.86)	50 (3.10)	13 (1.13)
Black or Black British	6,579 (1.60)	12 (1.64)	167 (10.35)	58 (5.05)
Chinese	1,446 (0.35)	≤5	6 (0.37)	≤5
Mixed	1,793 (0.44)	≤5	11 (0.68)	≤5
Other ethnic group	3,964 (0.96)	9 (1.23)	37 (2.29)	15 (1.31)
White	387,309 (94.18)	681 (92.91)	1,336 (82.83)	1,052 (91.64)

Values are mean ± SD or n (%). For groups with n ≤5 carriers, exact numbers cannot be shown caused by restrictions on reporting data from small samples. Self-reported ancestry categories exclude "prefer not to answer" and "do not know."

HDL-C = high-density lipoprotein cholesterol; LDL-C = low-density lipoprotein cholesterol.

from AoU (39.5% men; 18,360 with VTE) (**Table 2**) were included. Across both cohorts, 6,315 individuals carried at least 1 rare protein-disrupting variant in *LDLR*, *APOB*, or *PCSK9* (3,500 in UKB and 2,815 in AoU). *APOB* accounted for the largest proportion of carriers, followed by *PCSK9* and *LDLR*, with broadly comparable age and sex distributions between cohorts.

The prevalence of major cardiometabolic comorbidities was similar between variant carriers and noncarriers and reflected the underlying cohort populations. In UKB, hypertension affected 39% to 44% of participants, type 2 diabetes 9% to 12%, and coronary artery disease 4% to 14% (**Table 1**), with comparable frequencies observed in AoU (hypertension 41%-43%; type 2 diabetes 17%-20%; coronary artery disease 7%-12%) (**Table 2**). The distribution of VTE and aortic stenosis at baseline was likewise similar across carrier groups. Comprehensive characteristics for each gene and variant class are provided in **Supplemental Tables 10 and 11**.

ASSOCIATIONS OF RARE *LDLR*, *APOB*, AND *PCSK9* VARIANTS WITH LDL-C. Across both UKB and AoU, rare tv and AM variants in *LDLR*, *APOB*, and *PCSK9* produced the expected, strongly gene-specific effects

on LDL-C concentrations (**Figure 1**). In UKB, *LDLR* truncating variant carriers had markedly higher LDL-C (mean 225.1 mg/dL), corresponding to an adjusted increase of 77.6 mg/dL (95% CI: 70.9-84.4 mg/dL; ≈ 52.5%) compared with noncarriers in linear regression models. In contrast, *APOB* truncating variant carriers had substantially lower LDL-C (mean 76.8 mg/dL), corresponding to an adjusted decrease of 71.4 mg/dL (95% CI: 68.6-74.3 mg/dL; ≈ 48.3%) relative to noncarriers. *PCSK9* truncating variants were associated with more moderate LDL-C lowering (mean ≈ 109.7 mg/dL), corresponding to an adjusted decrease of 37.1 mg/dL (95% CI: 34.2-40.0 mg/dL; ≈ 25.1%) compared with noncarriers. Damaging missense variants exhibited the same overall directional pattern at the aggregate gene-burden level, but with generally attenuated effect sizes compared with truncating variants, although substantial functional heterogeneity likely exists at the individual variant level. For example, the *LDLR*-AM increase (55.7 mg/dL) was smaller than that observed for truncating variants (77.6 mg/dL), and the *APOB*-AM reduction (13.3 mg/dL) was markedly less pronounced than the 71.4 mg/dL decrease seen with truncating variants.

TABLE 2 Characteristics of the All of Us Study Participants by Rare Variant Carrier Status at Study Enrollment

	Noncarriers (n = 274,573)	LDLR (n = 699)	APOB (n = 1,240)	PCSK9 Rare (n = 876)
Age at recruitment, y	52.68 ± 16.94	51.39 ± 16.43	52.43 ± 16.69	51.30 ± 16.66
Female	166,210 (60.53)	420 (60.09)	747 (60.24)	525 (59.93)
BMI, kg/m ²	29.77 ± 7.52	29.88 ± 7.42	29.75 ± 7.44	30.08 ± 7.46
Lipid measurements				
Uncorrected LDL-C, mg/dL	98.69 ± 35.41	134.38 ± 60.92	81.89 ± 37.02	74.42 ± 29.69
Corrected LDL-C, mg/dL	108.01 ± 39.23	156.09 ± 72.75	88.98 ± 40.63	78.57 ± 31.08
HDL-C, mg/dL	54.66 ± 17.41	53.19 ± 15.98	57.34 ± 18.62	54.92 ± 17.58
Triglycerides, mg/dL	124.79 ± 85.88	113.57 ± 79.01	104.52 ± 71.21	114.54 ± 75.05
Total cholesterol, mg/dL	177.11 ± 42.82	206.29 ± 65.02	159.69 ± 46.18	151.16 ± 39.47
Disease				
Venous thromboembolism	17,763 (6.47)	40 (5.72)	87 (7.02)	39 (4.45)
Coronary artery disease	19,164 (6.98)	81 (11.59)	97 (7.82)	34 (3.88)
Aortic stenosis	1,725 (0.63)	≤20	≤20	≤20
Hypertension	112,518 (41.40)	291 (42.17)	519 (42.30)	354 (40.83)
Type 1 diabetes	4,713 (1.72)	≤20	≤20	≤20
Type 2 diabetes	48,773 (17.76)	117 (16.74)	257 (20.73)	175 (19.98)
Self-reported ancestry				
American Indian or Alaska Native	3,582 (1.30)	≤20	≤20	≤20
Asian	8,791 (3.20)	21 (3.00)	54 (4.35)	22 (2.51)
Black or African American	45,133 (16.44)	122 (17.45)	326 (26.29)	356 (40.64)
Middle Eastern or North African	1,653 (0.60)	≤20	≤20	≤20
More than 1 population	10,601 (3.86)	23 (3.29)	51 (4.11)	39 (4.45)
White	155,154 (56.51)	390 (55.79)	598 (48.23)	275 (31.39)
Native Hawaiian or Other Pacific Islander	250 (0.09)	NA	≤20	NA

Values are mean ± SD or n (%). For groups with n ≤20 carriers, exact numbers cannot be shown caused by restrictions on reporting data from small samples. Self-reported ancestry categories exclude "I prefer not to answer," "none indicated," "none of these," and "skip."

Abbreviations as in [Table 1](#).

Highly concordant patterns were observed in AoU. *LDLR* truncating variant carriers again showed elevated LDL-C (mean 154.9 mg/dL), corresponding to an adjusted increase of 45.5 mg/dL (95% CI: 35.9–55.1 mg/dL), whereas *APOB* truncating variants (mean 51.6 mg/dL) and rare *PCSK9* variants (mean 77.6 mg/dL) were associated with adjusted decreases of 57.3 mg/dL (95% CI: 51.2–63.4 mg/dL) and 31.0 mg/dL (95% CI: 25.9–36.1 mg/dL), respectively. Damaging missense variants demonstrated directionally concordant but generally smaller effects, with *LDLR*-AM associated with a 47.8 mg/dL increase and *APOB*-AM with a 7.7 mg/dL reduction, while *PCSK9*-AM effects (27.3 mg/dL reduction) were closer in magnitude to those of truncating variants. Regression estimates displayed consistent directions and magnitudes across cohorts at the aggregate gene-burden level, despite expected heterogeneity at the individual variant level, reinforcing the well-established functional effects of these genes on LDL metabolism ([Supplemental Tables 10 and 11](#)).

The common *PCSK9* R46L variant also demonstrated a clear allele-dosage relationship in both

cohorts, with LDL-C progressively lower among heterozygotes and homozygotes ([Supplemental Figure 1](#)), replicating its known functional impact.

ESTABLISHED THROMBOPHILIA VARIANTS SHOW ASSOCIATIONS CONSISTENT WITH PRIOR REPORTS. To evaluate the performance of the analytic framework in detecting thrombotic risk associations, we analyzed established thrombophilia genes and variants, including rare variants in *SERPINC1*, *PROC*, and *PROS1*, as well as the common prothrombotic variants rs6025 (factor V Leiden) and rs1799963 (prothrombin G20210A),²⁶ as positive controls. In the present study, carriers of ≥1 high-risk rare variant in *SERPINC1*, *PROC*, or *PROS1* were significantly associated with increased VTE risk in both cohorts (UKB: OR: 1.34; 95% CI: 1.24–1.44; $P = 1.68 \times 10^{-14}$; AoU: OR: 1.31; 95% CI: 1.22–1.40; $P = 1.91 \times 10^{-13}$), consistent in direction and magnitude with the original report (OR: 1.71; 95% CI: 1.43–2.03; $P = 1.40 \times 10^{-9}$).²⁶ In contrast, carriers of ≥1 low-risk rare variant were not associated with VTE. Factor V Leiden (rs6025) and prothrombin G20210A (rs1799963) were associated with VTE in both cohorts ([Supplemental Table 12](#)).

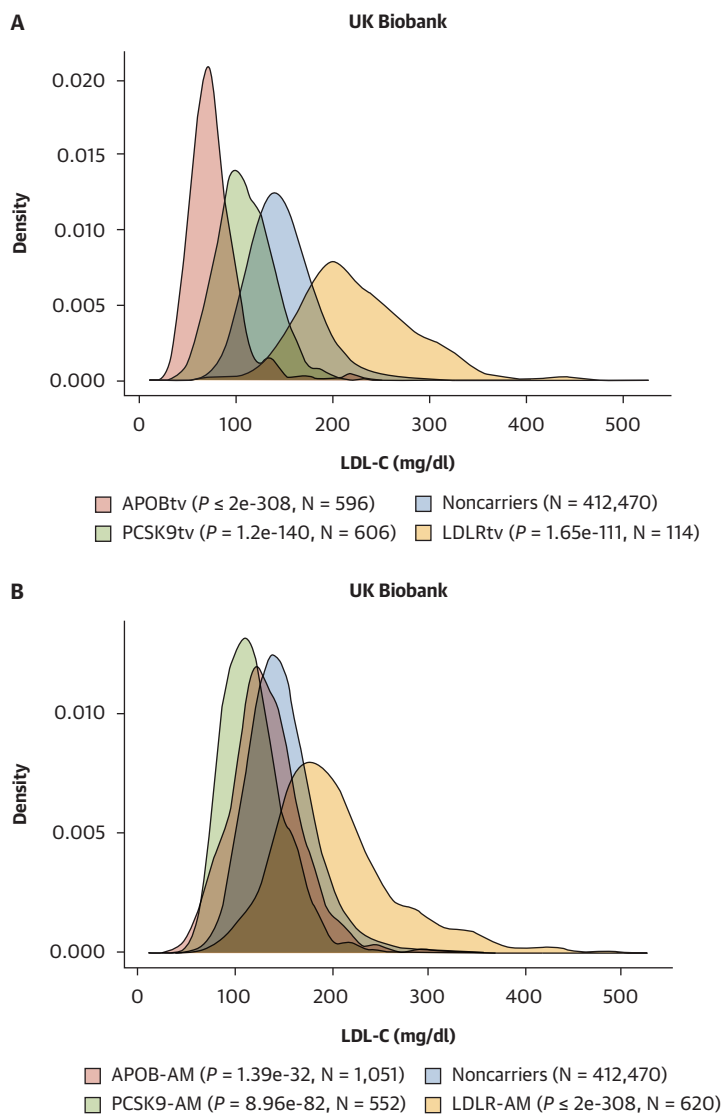
STATISTICAL POWER ESTIMATES INDICATE ABILITY TO DETECT MODERATE BUT NOT SMALL EFFECTS. Statistical power to detect protective associations among APOB, PCSK9 rare-variant carrier status, and VTE was estimated in the pooled UKB and AoU cohorts under a logistic regression framework (Supplemental Figure 2). At a 2-sided α level of 0.05, the study achieved 80% power to detect ORs of 0.76 for truncating variants only, 0.79 for annotated missense variants only, and 0.83 for the combined truncating-plus-annotated missense burden.

GENETIC LDL-C-LOWERING VARIANTS AND RISK OF VTE. Across UKB and AoU, rare protein-disrupting variants in *LDLR*, *APOB*, and *PCSK9* were not consistently associated with VTE risk (Figures 2A and 2B). For truncating variants, VTE ORs in UKB were 0.79 (95% CI: 0.31-2.04; $P = 0.63$) for *LDLR*tv, 0.88 (95% CI: 0.59-1.30; $P = 0.51$) for *APOB*tv, and 0.99 (95% CI: 0.68-1.43; $P = 0.95$) for *PCSK9*tv. In AoU, the corresponding estimates were 0.43 (95% CI: 0.15-1.24; $P = 0.12$) for *LDLR*tv, 1.05 (95% CI: 0.68-1.62; $P = 0.82$) for *APOB*tv, and 0.66 (95% CI: 0.44-0.98; $P = 0.04$) for *PCSK9*tv (Supplemental Table 13). Although individual point estimates varied in magnitude, nearly all CIs crossed 1, and the only nominally significant result was *PCSK9*tv in AoU. A similar pattern was observed for damaging missense burdens, in UKB, AM variants in *LDLR*, *APOB*, and *PCSK9* yielded ORs between 0.85 and 1.13, while the same burdens in AoU ranged from 0.71 to 1.10 (Supplemental Table 13).

Combined variant burdens were examined across both cohorts (Supplemental Table 13). When truncating and damaging missense variants were combined within or across genes, the *PCSK9* rare burden in AoU was associated with VTE (OR: 0.67; 95% CI: 0.49-0.92; $P = 0.01$), whereas the corresponding estimate in UKB was 0.92 (95% CI: 0.69-1.21; $P = 0.53$). The *PCSK9* R46L variant was associated with OR: 1.08 (95% CI: 1.01-1.17; $P = 0.03$) in UKB and OR: 1.02 (95% CI: 0.93-1.13; $P = 0.64$) in AoU.

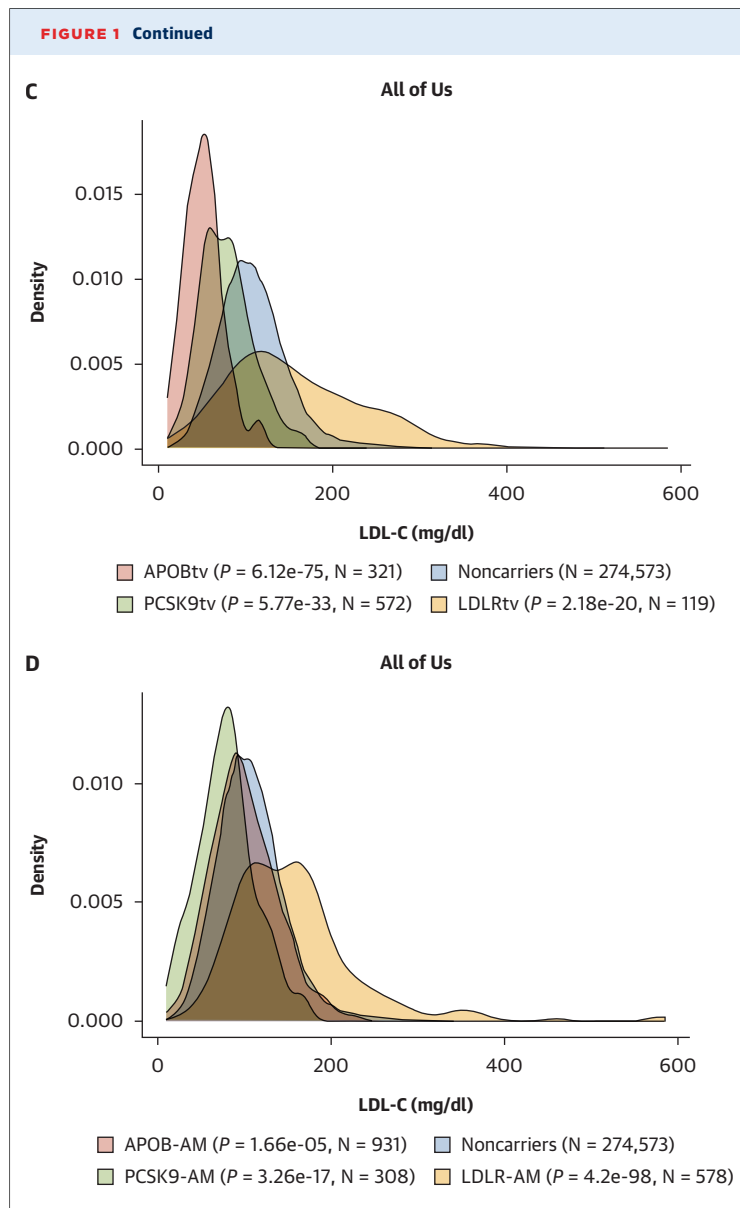
In meta-analysis of UKB and AoU (Figure 2C), ORs for rare variant burdens in *LDLR* ranged from 0.60 (95% CI: 0.30-1.22; $P = 0.16$) for truncating variants to 0.98 (95% CI: 0.78-1.23; $P = 0.85$) for the combined burden. For *APOB*, estimates ranged from 0.95 (95% CI: 0.71-1.28; $P = 0.75$) to 1.07 (95% CI: 0.89-1.29; $P = 0.46$) for the damaging missense variants. For *PCSK9*, pooled ORs were 0.82 (95% CI: 0.62-1.08; $P = 0.16$) for truncating variants, 0.79 (95% CI: 0.57-

FIGURE 1 LDL-C Distributions in Carriers of Rare Truncating Variants or Damaging Variants in *LDLR*, *APOB*, and *PCSK9* in the UK Biobank and All of Us



(A) Kernel density distributions of directly measured low-density lipoprotein cholesterol (LDL-C) at the UK Biobank baseline visit for participants carrying predicted truncating variants (tv) in *APOB* (red), *PCSK9* (green), or *LDLR* (orange), and for noncarriers of predicted truncating variants in these genes (blue). Statistical differences between variant carriers and noncarriers were assessed using linear regression adjusted for age, age², sex, genotyping array, and the first 10 genetic principal components. (B) LDL-C distributions for carriers of variants predicted to be damaged by AlphaMissense (AM) in *APOB*, *PCSK9*, or *LDLR* in UK Biobank. (C and D) Parallel analyses in the All of Us Research Program for truncating variants (C) and AlphaMissense-predicted damaging variants (D). Regression models were adjusted for age, age², sex, and the first 10 genetic principal components; the genotyping array was not included as a covariate in All of Us.

Continued on the next page



1.10; $P = 0.16$) for damaging missense variants, and 0.80 (95% CI: 0.65-0.99; $P = 0.04$) for the combined burden. The additive model for the common *PCSK9* R46L variant yielded an OR of 1.06 (95% CI: 1.00-1.12; $P = 0.06$).

SENSITIVITY ANALYSES. Sensitivity analyses using additional cardiovascular phenotypes reproduced well-established gene-disease relationships. As expected, *LDLR* rare variants were strongly associated with CAD and AS, with markedly elevated risks for both truncating and damaging missense carriers (Supplemental Tables 14 and 15). These findings align with the known biological effects of *LDLR* loss

of function on atherosclerotic and valvular disease. Analyses were repeated after excluding individuals with CAD. Effect estimates for rare variant burdens in *LDLR*, *APOB*, and *PCSK9* were similar in magnitude to those observed in the primary analysis. *PCSK9* R46L variant reached nominal significance in selected UKB models after excluding individuals with CAD, whereas corresponding associations in AoU remained nonsignificant. In meta-analysis, the observed direction of association was opposite to the expected protective effect of LDL-C lowering, and findings were not consistently observed across cohorts or primary analyses (Supplemental Table 16, Supplemental Figure 3). Sensitivity analyses using alternative validated VTE definitions yielded broadly consistent effect estimates, supporting the robustness of the primary findings despite differences in phenotype ascertainment stringency (Supplemental Figure 4).

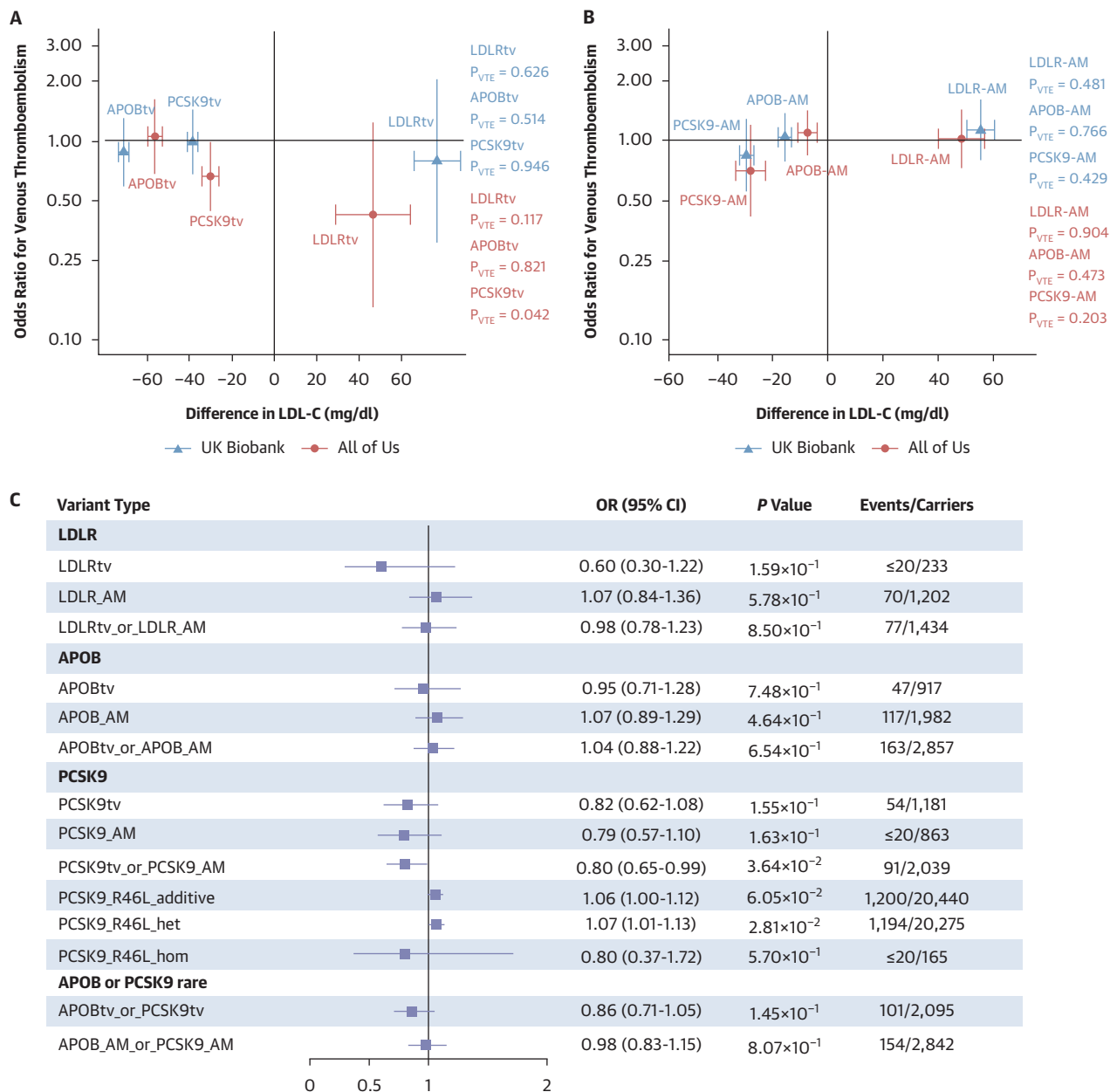
ADDITIONAL COMMON VARIANT AND MULTIVARIABLE MR ANALYSES. To further contextualize the rare variant findings, we performed complementary common variant analyses in European ancestry participants from the AoU cohort.

Baseline characteristics were compared between individuals in the top quintile of an LDL-C polygenic risk score (PRS; $n = 56,525$) and rare coding variant carriers across *LDLR*, *APOB*, and *PCSK9* ($n = 2,253$). High-PRS individuals had higher mean triglycerides compared with rare variant carriers (129.98 mg/dL vs 107.81 mg/dL; $P < 0.001$), despite comparable BMI (29.73 kg/m² vs 29.91 kg/m²). LDL-C levels differed substantially across gene-specific rare variant groups in the expected directions (*LDLR* carriers: 134.29 mg/dL; *APOB* carriers: 78.57 mg/dL; *PCSK9* carriers: 72.11 mg/dL) (Supplemental Tables 17 and 18).

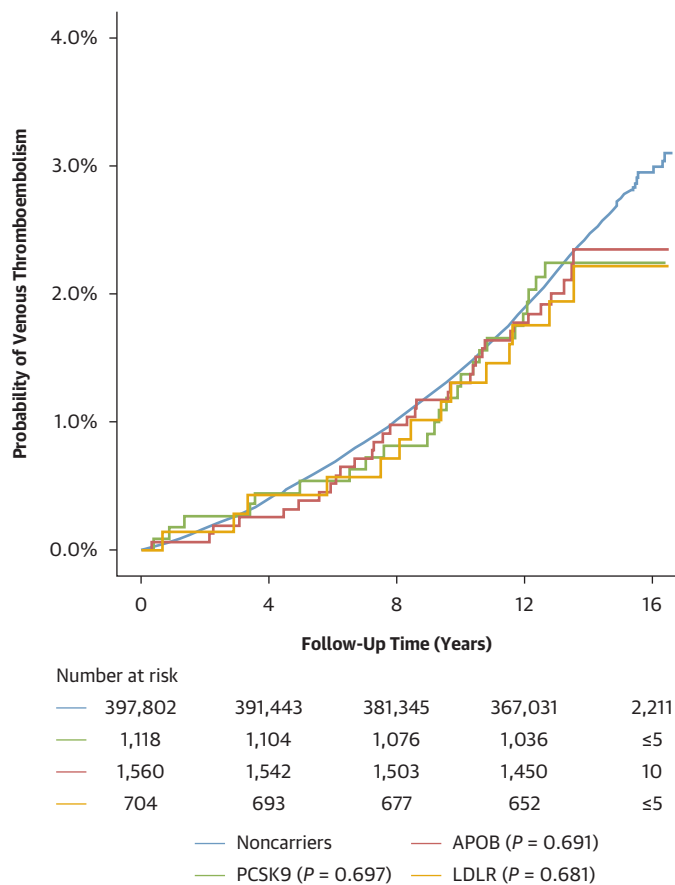
In multivariable MR jointly modeling LDL-C, HDL-C, and triglycerides, LDL-C was nominally positively associated with VTE (OR: 1.09; 95% CI: 1.01-1.17; $P = 0.02$), consistent with prior reports²⁴ (Supplemental Figure 5).

In multivariable MR jointly modeling LDL-C and BMI, the LDL-C association was attenuated to non-significance (OR: 1.08, 95% CI: 0.99-1.17; $P = 0.07$), whereas BMI remained independently associated with VTE (OR: 1.58; 95% CI: 1.26-1.99; $P < 0.001$) (Supplemental Figure 6).

PROSPECTIVE ANALYSES OF INCIDENT VTE. We evaluated whether LDL-modifying genetic variants influenced incident VTE in UKB using prospective survival analyses. Kaplan-Meier curves for rare variant burdens in *LDLR*, *APOB*, and *PCSK9* were nearly identical to those of noncarriers, with no

FIGURE 2 LDL-C Differences and Venous Thromboembolism Associations for Rare and Common Variants in *LDLR*, *APOB*, and *PCSK9*

(A and B) Differences in LDL-C (x-axis) and corresponding ORs for venous thromboembolism (y-axis) among carriers of *LDLR*, *APOB*, and *PCSK9* rare protein-tv, damaging AMs in UK Biobank (blue) and All of Us (red). Points represent mean LDL-C differences relative to noncarriers, with the direction of LDL-C effects differing across genes, and vertical error bars indicate 95% CIs for the VTE association. ORs for venous thromboembolism were estimated using the Firth logistic regression with variant carrier status, age, age², sex, genotyping array (only in UK Biobank), and the first 10 genomic principal components as independent variables. (C) Forest plot depicting the association of rare variants in *LDLR*, *APOB*, and *PCSK9*, as well as the common *PCSK9* R46L variant, with risk of venous thromboembolism. ORs and 95% CIs were obtained from meta-analyses of the UK Biobank and All of Us cohorts. Variant groups include gene-specific burdens of predicted tv, AMs, combined tv+AM burdens, and cross-gene aggregated burdens, as well as additive, heterozygous, and homozygous models for the *PCSK9* R46L variant. Squares represent point estimates for the ORs, and horizontal lines indicate 95% CIs; carrier counts for each group are shown on the right. For groups with $n \leq 20$ carriers, exact numbers cannot be shown because of restrictions on reporting data from small samples. Abbreviations as in Figure 1.

FIGURE 3 Probability of Incident Venous Thromboembolism in Carriers of LDL-Modifying Rare Variants in UK Biobank

The figure depicts the probability of incident venous thromboembolism during up to 16 years of follow-up among UK Biobank participants who had no history of VTE at enrollment and carried rare LDL-modifying variants in *LDLR*, *APOB*, or *PCSK9*, compared with noncarriers. Curves show the cumulative incidence for carriers of rare variants in *PCSK9* (green), *APOB* (red), and *LDLR* (yellow), and for noncarriers (blue). P values were derived from Cox proportional hazards models, comparing each variant group separately with noncarriers and adjusting for sex, age, age², the first 10 principal components, and genotyping array. For groups with $n \leq 5$ carriers, exact numbers cannot be shown because of restrictions on reporting data from small samples. Abbreviations as in Figure 1.

separation over more than 16 years of follow-up (Figure 3). Log-rank tests likewise showed no evidence of differences in cumulative VTE incidence across any gene-specific burden group.

Because the *PCSK9* p.R46L variant confers a large lifelong reduction in LDL-C, we additionally examined incident VTE among heterozygous and homozygous carriers. Both groups displayed incidence curves that closely overlapped with noncarriers, and log-rank statistics were nonsignificant (Supplemental Figure 7).

DISCUSSION

In this large 2-cohort genetic analysis of UKB and AoU, we did not find consistent evidence that life-long genetic perturbation of LDL-C through rare functional variation in *LDLR*, *APOB*, or *PCSK9* has a moderate or large effect on VTE risk (Central Illustration), although smaller effects (eg, OR: 0.85–0.95) cannot be excluded. Although these variants produced substantial, gene-specific differences in circulating LDL-C, corresponding effect estimates for VTE were modest and directionally heterogeneous across genes and cohorts. Some variability in effect estimates was observed across cohorts. This may reflect differences in phenotype definition and ascertainment, population characteristics, and baseline risk profiles between UKB and AoU. In addition, given the rarity of the variants studied, effect estimates may be sensitive to statistical variability. Despite these differences, the overall pattern of results was consistent in not supporting a moderate or large effect of genetically mediated LDL-C differences on VTE risk.

These findings are further contextualized by complementary common variant analyses performed in AoU European ancestry participants. Replicating the multivariable MR framework described by Klarin et al,²⁴ using common variant instruments, LDL-C remained nominally associated with VTE after jointly modeling HDL-C and triglycerides, consistent with prior reports. However, when BMI was additionally incorporated into the multivariable MR framework, the LDL-C association was attenuated to nonsignificance, whereas BMI remained strongly and independently associated with VTE risk. Furthermore, individuals with high polygenic LDL-C risk exhibited broader metabolic differences compared with rare coding variant carriers, whose lipid perturbations were more pathway-specific. Prior epidemiologic and mechanistic studies have also linked obesity, metabolic syndrome, and inflammatory pathways with VTE risk, whereas direct associations between LDL-C itself and VTE have been less consistent.^{31,32} Collectively, these findings suggest that previously reported common variant associations between LDL-C genetic instruments and VTE may partly reflect broader metabolic and obesity-related pathways rather than a direct causal effect of LDL-C itself. These observations also help reconcile the apparent discordance between rare and common variant approaches: rare coding variants in *LDLR*, *APOB*, and *PCSK9* provide relatively specific perturbations of canonical LDL-C biology, whereas

CENTRAL ILLUSTRATION Genetically Mediated LDL-C and Venous Thromboembolism Risk

Lifelong genetically mediated differences in LDL cholesterol (LDL-C) were not consistently associated with venous thromboembolism (VTE) risk, suggesting that LDL-C is unlikely to be a major causal determinant of VTE.

STUDY DESIGN

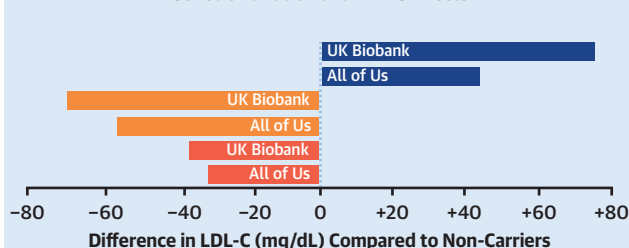
Examined rare functional variants in LDL-related genes (*LDLR*, *APOB*, and *PCSK9*) and the common *PCSK9* R46L variant



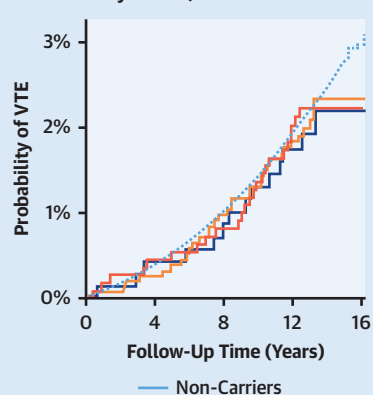
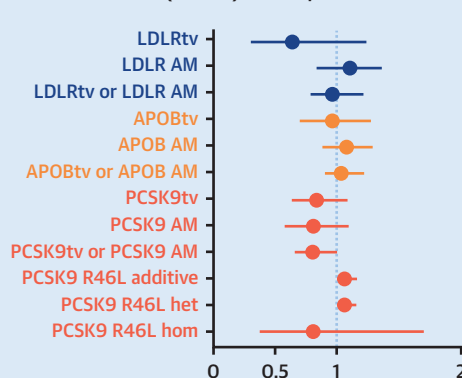
Combined analysis of two large population-based cohorts (UK Biobank and All of Us), totaling >700,000 participants



Included cross-sectional analyses, meta-analysis across cohorts, and prospective time-to-event analyses for incident VTE

RESULTS**Genetic Variation and LDL-C Effects****LEGEND**

■ / LDLR variants
 ■ / APOB variants
 ■ / PCSK9 variants
 tv = truncating variant
 AM = Alpha Missense predicted damaging variants
 het = heterozygous
 hom = homozygous

Genetic Variation and VTE Outcomes**Probability of VTE, UK Biobank Cohort****Odds Ratios (95% CI) for VTE, Both Cohorts****Conclusions and clinical implications**

- Despite large LDL changes, VTE risk remains largely unchanged.
- Lowering LDL-C alone is unlikely to meaningfully reduce VTE risk.
- Any VTE benefit observed in clinical trials of lipid-lowering therapies may involve mechanisms beyond LDL-C reduction alone.

Sui Y, et al. JACC. 2026;■(■):■-■.

Rare functional variants in *LDLR*, *APOB*, and *PCSK9* were used to evaluate whether lifelong genetic differences in low-density lipoprotein cholesterol (LDL-C) influence venous thromboembolism (VTE) risk. Across 2 large biobanks, these variants produced substantial differences in LDL-C but were not consistently associated with VTE in cross-sectional or prospective analyses. These findings suggest that low-density lipoprotein-related pathways may not substantially influence venous thrombosis risk.

common variant polygenic instruments capture a broader set of metabolic and pleiotropic mechanisms that co-vary with LDL-C levels.

Indeed, lipid-lowering agents, particularly statins, exert a broad spectrum of pleiotropic, non-LDL-mediated effects, including anti-inflammatory, endothelial, and antithrombotic actions.³³ Experimental studies have also implicated LDL receptor pathways in coagulation and endothelial regulation, including modulation of factor VIII levels.^{34,35} A recent perspective emphasized that statins, PCSK9 inhibitors, and other lipid-lowering drugs may attenuate thrombotic risk through pathways affecting tissue factor expression, thrombin generation, platelet activation, and endothelial function.³⁶ However, these drug-specific and context-dependent mechanisms are not directly captured by inherited perturbations in *LDLR*, *APOB*, or *PCSK9*, which primarily influence circulating LDL-C concentrations. Similar uncertainty has been observed for other cardiometabolic therapies, including GLP-1 receptor agonists, for which recent meta-analyses have reported inconsistent associations with VTE risk.^{37,38} Together, these considerations suggest that any VTE benefit observed in randomized trials may not be attributable to LDL-C reduction alone. Future studies should further evaluate whether specific lipid-lowering therapies influence VTE risk through broader metabolic, inflammatory, endothelial, or coagulation-related pathways independent of LDL-C lowering itself.

Whether PCSK9-related pathways influence VTE risk through mechanisms independent of LDL-C lowering remains an open question. In the present study, PCSK9 rare variant burdens showed a modest but somewhat more consistent directional pattern toward VTE protection compared with *LDLR* and *APOB* variants, particularly in AoU and in the combined meta-analysis; however, this signal was not consistent across cohorts and did not reach conventional significance thresholds in UKB.

In addition to its canonical role in LDL metabolism, PCSK9 has been implicated in inflammatory processes, endothelial function, and thrombosis-related pathways,^{39,40} which may differ from the primary mechanisms of *LDLR* and *APOB* variants. However, these potential mechanisms remain incompletely understood and do not establish a causal relationship with VTE. Whether such LDL-independent effects contribute to any differential association with VTE warrants further investigation in larger studies.

Importantly, the analytic framework identified established thrombophilia rare variants with effect sizes comparable to a prior report,²⁶ supporting the validity of the study design. In addition, the

precision of the pooled estimates suggests that moderate associations of a magnitude typically considered clinically meaningful would likely have been detected. While small effects cannot be excluded, the overall pattern does not support a substantial influence of genetically mediated LDL-C perturbation on VTE risk.

From a clinical perspective, these results suggest that targeting LDL-C alone is unlikely to be an effective strategy for VTE prevention. Although lipid-lowering therapies may confer modest reductions in VTE risk, our findings support the interpretation that such effects, if present, are unlikely to be mediated primarily through LDL-C lowering. Future studies should examine whether specific lipid-lowering therapies confer VTE benefit through LDL-independent mechanisms and whether such effects vary across drug classes.

This study has several strengths. It leverages 2 large, deeply phenotyped population-based biobanks with harmonized genetic and phenotypic data, enabling robust cross-cohort evaluation. The genetic instruments demonstrated strong, gene-specific effects on LDL-C and recapitulated known associations with coronary artery disease, supporting biological validity. The use of complementary analytic approaches, including case-control, time-to-event, and sensitivity analyses, provides a comprehensive assessment of the relationship between genetically mediated LDL-C and VTE.

STUDY LIMITATIONS. Anticoagulation therapy could not be consistently incorporated caused by limited medication data. Despite large overall sample sizes, statistical power remains limited for ultra-rare variants, particularly within gene-specific truncating burdens. Differences in phenotype definitions and event rates across cohorts may contribute to heterogeneity. Although findings were broadly consistent across multiple validated VTE phenotype definitions, some degree of phenotype misclassification remains possible in EHR-based studies. In addition, incident VTE analyses were not performed in AoU because of limited follow-up duration. VTE phenotypes did not distinguish between provoked and unprovoked events, which may represent biologically distinct subtypes. Our study may also be subject to healthy volunteer or survivor bias inherent to biobank-based designs. However, sensitivity analyses excluding individuals with coronary artery disease yielded broadly consistent results, suggesting that such bias is unlikely to materially affect the primary conclusions. Finally, genetic analyses reflect lifelong exposure and may not fully capture the acute or pleiotropic effects of pharmacologic lipid-lowering therapies.

CONCLUSIONS

We found across 2 large population-based biobanks, rare truncating and damaging missense variants in *LDLR*, *APOB*, and *PCSK9* were associated with substantial, lifelong differences in LDL-C but were not consistently associated with VTE risk. These findings suggest that genetically mediated perturbations of LDL-C are unlikely to exert a major influence on venous thrombosis at the population level. Complementary common variant and multivariable MR analyses further suggest that previously reported LDL-C associations with VTE may partly reflect broader metabolic and obesity-related pathways rather than a direct causal effect of LDL-C itself. In contrast to pharmacologic studies reporting modest VTE reductions with certain lipid-lowering regimens, our findings support the interpretation that any potential VTE benefit of lipid-lowering therapy may not be attributable to LDL-C reduction alone. Future investigations should clarify whether specific lipid-lowering agents influence VTE risk through LDL-independent metabolic, inflammatory, endothelial, or coagulation-related mechanisms.

DATA AVAILABILITY Data from UKB are available to approved researchers through application to the UKB Access Management System (www.ukbiobank.ac.uk), under application number 17488. Data from the AoU Research Program are accessible to researchers through the AoU Researcher Workbench (www.researchallofus.org) under an approved data use agreement. Individual-level genomic and phenotypic data cannot be publicly shared caused by participant privacy and program policies.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: Lifelong genetically mediated differences in LDL-C are not consistently associated with VTE risk, suggesting that previously reported lipid-related associations with VTE may partly reflect broader metabolic or pleiotropic pathways rather than LDL-C itself.

TRANSLATIONAL OUTLOOK: Future studies should determine whether specific lipid-lowering therapies reduce VTE risk through broader metabolic, inflammatory, endothelial, or coagulation-related pathways independent of LDL-C lowering itself and further clarify the biological mechanisms linking lipid metabolism and thrombosis.

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APPENDIX For supplemental figures and tables, please see the online version of this paper.